

System Architecture Blueprint

Document Link:

<https://docs.google.com/document/d/1JQYO9HGbgO-ggfqeEMzwpT49bIVluM3oXfHQziNsfUs/edit?usp=sharing>

A comprehensive technical architecture document covering:

- **Executive Overview**
 - Platform vision and strategic positioning
- **High-Level System Architecture**
 - 6-layer architecture (Frontend, API, Business Logic, Realtime, Database, External Services)
- **User Roles & Permission Matrix**
 - 5 roles with granular RBAC definitions
- **Database Architecture**
 - Full relational schema with **trucks** and **dispatches** tables, column types, constraints, and indexing strategy
- **Realtime Operational Flow**
 - 9-step pipeline from fleet update to analytics recording
- **Realtime Infrastructure**
 - Socket.IO, Redis Pub/Sub, Kafka, and standardized event taxonomy
- **API Architecture**

- Complete REST API specifications for Auth, Trucks, Dispatches, Tracking, and Analytics

- Security Architecture

- 10 security layers including JWT, RBAC, multi-tenant isolation, RLS, encryption, and disaster recovery

- Scalability Strategy

- 4-phase scaling roadmap with horizontal scaling components

- Pain Points & Solutions

- 12 operational problems with platform solutions

- Future Expansion

- 11 AI-powered and advanced features

- Final System Vision

- 4-stage evolution to regional logistics network